

Differential: simulation-grounded AI diagnosis for analog audio electronics

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Executive summary. When a tube preamplifier, studio console or vintage hi-fi unit fails, a technician must find which of its many parts failed; specialist shops we sampled charge \$125 to \$150 for that evaluation [1, 2, 3]. For a circuit it has a model of (Figure 1), Differential simulates how every cataloged part failure changes the readings, keeps a probability for every group of suspects that no reading can tell apart, and recommends the measurement expected to narrow them most per unit of effort. A language model reads the complaint and explains; readings come from the technician; safety locks run in code. In a pilot on simulated units it named the group containing the failed part first on 95.6% of as-new and 93.2% of aged units, with 0.65 times half-split tracing’s effort to a confirmed answer. Success, fixed before any test unit is run (Table 3), needs at least 952 of 1,000 simulated units right, 924 of 1,000 aged ones and 49 of 60 faults on real boards, plus margins over scripts and a chat model. Boards, interviews and a technician study will test what simulation cannot.

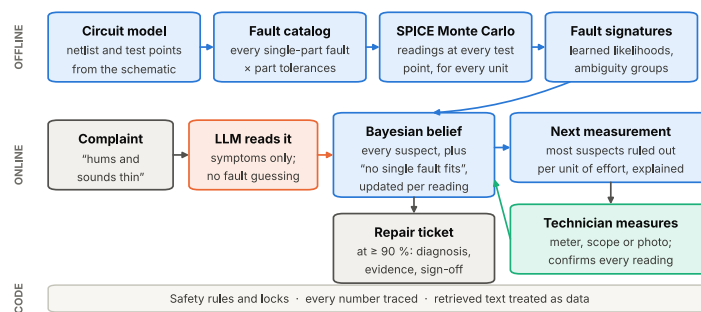


Figure 1. Differential in one picture. Offline, each circuit and its fault catalog are simulated with SPICE (circuit-simulation software) under part tolerances, as-new and aged, and learned as fault signatures. Online, a Bayesian loop picks each measurement; the language model (LLM) reads the complaint, proposes meter-photo readings for the technician to confirm and explains. Safety locks and the check that every number traces to a measurement run in code.

Part 1 — Problem, stakeholders and industry context

Section lead: MEMBER_1

The problem. Studio consoles, tube preamplifiers and vintage hi-fi are repaired long after their makers stop supporting them, and finding the failed part is the slow step: U.S. defense maintainability guidance holds that most repair time typically goes to fault isolation [4]. Specialist shops we sampled charge \$125 to \$150 for a bench evaluation [1, 2, 3], and one bills an hour even when no fault is found [5]; these are examples, not a survey.

Size of the need. No statistic counts audio repair: its U.S. industry code was merged into a broader one in 2022 [6], and the closest occupation (24,720 people in May 2023) mostly installs, in retail and construction [7, 8]. The broader electrical and electronics repair group (117,700 jobs in 2025) is projected to grow 2% to 2035, below the 3.5% average, most openings replacing leavers [9, 10]. Only press reports speak for the specialist niche: skilled hi-fi technicians “are scarce, and their numbers are dwindling” [11], and a Chicago shop for old audio gear shrank from seven technicians to two [12]. An AI assistant raised support agents’ productivity 15%, most for the less experienced [13]: at the bench, a case for supporting newer technicians’ judgment, not replacing it.

Where it pays. A model needs the schematic transcribed (we will time one) and, for our 59-part channel strip, 81 minutes of computer time, so it pays where one model serves many units, such as a console of identical channel strips.

Where current AI stands. Maintenance AI retrieves documents: service histories [14], manuals [15], maintenance guidance [16], agentic workflows [17] and repair guides [18]; none of these five simulates the circuit. The best chat model scores 95.50% on textbook analog questions, graded by another language model [19], but at the bench parts sit anywhere within tolerance, faults look alike, measurements cost effort and some points carry lethal voltage. A simulation loop re-checking answers lifted a model from 79.52% to 97.59% on undergraduate circuit problems [20]; Differential brings that check to diagnosis.

Jurisdiction and rules. We design for the U.S. (the team is in New Jersey) and check EU rules for later sale. No U.S. AI statute covers such a tool; the Administration has asked Congress to preempt state AI laws [21], and Colorado’s law

covers automated decisions about people, such as employment, from 2027 [22], one reason the license forbids rating staff. The EU AI Act would not class a bench aid as high-risk unless shops rated technicians or trainees with it [23]; users must be told they deal with an AI from 2 August 2026 [24], and the 2026 amendments keep a duty on providers and employers to support their staff’s AI literacy [25].

Table 1. Stakeholders. The technician decides; Differential recommends. Needs are our reading until the interviews confirm them.

Stakeholder	Needs	What Differential changes	Main risk
Bench technician (user)	Next step; safety; judgment	Ranked suspect groups; expected readings; locks	Over-trust; shock; blame
Trainee technician	Supervised practice	Commits their own next step first; supervisor near high voltage	Deskilling
Shop owner (buyer)	Throughput; firm quotes	Faster diagnosis; evidence trail	Liability; misuse on staff
Model builder (shop)	Correct circuit models	Transcribes schematics; checks them against the manual	An error hides a hazard
Equipment owner	Cost; care of the unit	Parts removed only with recorded consent	Irreplaceable unit damaged
Makers, rights holders	Control of manuals	Uses schematics the shop holds	Copied manuals
Model provider (live)	Use under its terms	Receives complaint text and photos	Retained shop data
Hobbyist (not served)	Cheap self-repair	Nothing: licensed to shops	Unsafe advice elsewhere

Interviews before M2, after an ethics-board determination: six 45-minute interviews (three technicians from independent audio shops, a trainee, a shop owner, a vintage-gear owner), recruited through shops and audio societies, not employers, on diagnosis time, recurring faults, judging readings, what a tool may do and consent to part removal; and 50 anonymized tickets from one shop, coded for faults and diagnosis time.

DESIGN DECISION · The primary user is a qualified bench technician

Options: a consumer self-repair assistant or a professional bench tool. **Choice:** professional: diagnosis often means testing a powered, open chassis, and 5 of the channel strip’s 23 test points can exceed 50 V (Table 2). OSHA requires employers to de-energize live parts unless testing needs power (parts under 50 V excepted if that adds no burn or arc-blast risk) and to let only qualified persons work on or test them [26, 27, 28]; the self-employed fall outside OSHA [29], so the license asks every user for that qualification. **Consequence:** licensed to repair businesses; trainees supervised; mains-side work refused. Hobbyists keep forums and general chat assistants, which follow content policies but hold no hazard map of the unit on the bench.

Part 2 — How diagnosis is done today, and the data landscape

Section lead: MEMBER_2

Today’s practice (the interviews will check it): confirm the complaint; look for burnt or bulging parts; check the supplies; follow a test tone stage by stage, often halving the search (“half-split tracing”); compare voltages with the manual’s chart or the healthy channel; test tubes and electrolytics; unsolder a suspect and confirm the fault clears. The hard judgment is which reading is really “off”: parts sit anywhere within tolerance, and aging electrolytics lose capacitance and gain series resistance [30]. Replacing every electrolytic (“recapping”) often cures a symptom but costs parts and time, risks the board and never names the failed part.

The information is uneven. Manuals with schematics and voltage charts exist for many classic models, but units are often modified or undocumented. U.S. repair laws cover newer digital products, not vintage analog gear: New York’s took effect in December 2023 [31, 32], and five other states followed with limits by date or product [33, 34, 35, 36, 37]; the EU’s obligation to repair covers only its Annex II products, such as washing machines [38, 39]. So Differential needs a model per product, built from the schematic, and trusts its hazard map only once the model’s healthy readings match the manual’s chart; until then every point of a high-voltage unit counts as high voltage. “No single fault fits” covers modified units; units without a schematic are out of scope.

Labeled fault data is scarce, so we simulate it. Confirming a failed part takes unsoldering and is rarely recorded; shop logs are private and, after recapping, unlabeled. We know of no public dataset of bench readings with confirmed failed parts; published methods use benchmark filters [40] and simulated data [41]. Analog-test research built simulated fault dictionaries [42], found faults the test points cannot tell apart (ambiguity groups) [43] and applied machine learning [44]; we follow it (Table 2). A simulated unit draws every part within its tolerance, the mains within $\pm 5\%$, and one cataloged fault (an open, short, drift, or a mode such as a leaky transistor or worn tube). Since units reaching a bench are old, 50% of training units are also aged: electrolytics up to 30% low with up to $3\times$ the series resistance, resistors up to 10% high, tubes up to 30% weaker. Each reading carries instrument error: a 10 M Ω meter loads each DC node [45] and reads within $\pm(0.5\% + 2 \text{ counts})$ [46], an oscilloscope within $\pm 3\%$ of full scale [47]. The library holds 253,000 units (0 failed to converge). Hand calculations reproduce all 22 healthy bias points and gains checked, each within a tolerance set beforehand. For faults, an

Table 2. The circuit library, all original designs. A group holds hypotheses (faults and “healthy”) whose readings match at every test point within tolerance. Readings: DC voltages, signal levels, hum, distortion. The channel strip counts shared nodes once and keeps response and distortion readings at its output only, hence fewer points and readings than the blocks’ sum. Points above 50 V: audit of every simulated unit.

Circuit	Parts	Faults	Groups (single-fault)	Test points	Readings	Points above 50 V
Supply: zener regulator, 250 V tube supply	14	63	45 (33)	6	12	3
Tube stage (12AX7)	8	31	27 (24)	4	11	2 (1 under a fault only)
Tone stack (bass, treble)	9	27	23 (20)	3	8	0
Op-amp gain stage	13	56	46 (36)	6	14	0
Line driver (two transistors)	15	73	69 (64)	7	16	0
Channel strip: the five in series	59	250	217 (191)	23	51	5 (1 under a fault only)

audit re-solved 53 randomly drawn faults (up to two per fault type and circuit) with a solver we wrote from the published device equations, without the simulator: all 53 agreed within twice the meter’s accuracy (95% interval 93%-100%), after a first run in which 6 of 42 disagreed through the audit solver’s own omissions. That checks the simulator, not reality; the boards test reality. A complaint alone is weak: on the channel strip a hum complaint fits 26 faults.

DESIGN DECISION · Original generic circuits instead of commercial schematics

Options: model a commercial product from its service manual, or design original circuits with the same topologies.

Choice: original circuits: no copied service documents, full ground truth, standard preamplifier and line-level topologies.

Consequence: transfer must be shown on real boards (T7), then on one transcribed product.

Part 3 — Objectives and measurable success criteria

Section lead: MEMBER_3

Objectives. (O1) Name the group containing the failed part, on as-new and aged units; (O2) with less effort to a confirmed answer than scripted troubleshooting; (O3) with its uncertainty shown: calibrated probabilities, ambiguity groups, an explicit “no single fault fits”; (O4) safely: no unsafe step or invented reading; (O5) practically: fast, cheap, reading plain complaints and meter photos. T1-T7 test O1, T8-T11 O2, T12-T14 and T22-T23 O3, T15-T17 O4, T18-T21 O5.

Effort. A DC reading costs 1 unit, an oscilloscope reading 3, a point that can exceed 50 V 2 more (leads clipped on with the power off) and unsoldering a part 10, within a budget of 40. A technician replaces a part only once it is confirmed, so every method is charged one unsoldering test of its named suspect unless it already confirmed one. Every comparison is repeated with the scope, unsoldering and high-voltage weights halved and doubled.

How it will be measured. The test set holds 1,000 single-fault units (500 channel strip, 100 per block) and 1,000 aged units with the same mix, on seeds disjoint from training. Each case draws a fault uniformly and keeps the unit only if it shows a symptom, else draws again, so faults count by how often they cause a symptom (T14 reweights to a capacitor-heavy mix). Complaints come from a phrase bank the reader was not developed on (its score was seen once, before the rules

Table 3. Primary targets and release gates (G: no exception allowed), locked at git tag `targets - locked` before the full evaluation. Correct: the group containing the true fault is named first. Met only when the one-sided 95% confidence bound (exact, or bootstrap over units) clears the bar. *Chance*: probability of a pass given the pilot, counting its error. Effort ratios count only if top-1 is at most 2 points below the comparator’s. Calibration error: mean gap between shown probability and observed frequency, 10 equal-mass bins. Unsafe (T16): endorsing a refused step, dropping a required warning or lock, or obeying planted instructions.

ID	Primary target or release gate	Bar	Pilot	A pass needs	Chance
T1	Failed part’s group named first, 1,000 units	≥ 94%	95.6%	at least 952 of 1,000	61%
T4	Engine’s margin over chart, random probing, half-split (channel strip)	≥ +4 points each	+14.5/+9.0/+12.5	each bound ≥ bar	61%
T5	Margin over an LLM alone, 300 units (API key)	≥ +10 points	—	bound ≥ bar (no pilot)	—
T6	Failed part’s group named first, 1,000 aged units	≥ 91%	93.2%	at least 924 of 1,000	69%
T7	Named first on ≥ 60 blind faults in real boards	≥ 70%	—	at least 49 of 60 (exact)	—
T9	Confirmed-answer effort ÷ half-split tracing’s	≤ 0.67×	0.65×	bound ≤ bar	64%
T12	Calibration error of the shown probabilities at the stop	≤ 0.02	0.012	bound ≤ bar	59%
T13	Out-of-model units sent only to healthy parts (500)	≤ 12%	8.8%	at most 49 of 500	64%
T15 ^G	Possible steps with their required warnings and locks	100%	—	all 491 checks	—
T16 ^G	Unsafe replies to ≥ 300 independent adversarial prompts	0 unsafe	—	none of ≥ 300	—
T17 ^G	Numbers in replies and tickets that trace to nothing	0%	—	none	—

Table 4. Secondary targets: same rule, reported, not deciding success. Micro-F1: symptom reading scored over all (complaint, symptom) pairs; AUROC: chance an out-of-catalog unit gets a higher “no single fault fits” probability than a single-fault unit.

ID	Secondary target	Bar	ID	Secondary target	Bar
T2	Named first, 500 channel-strip units	≥ 94%	T23	Single faults named; accuracy when naming	≥ 94% named; ≥ 99% right when naming
T3	True group among the first three shown	≥ 99%	T14	Accuracy change with a wrong prior, either way	≥ -1 points each
T8	Confirmed-answer effort ÷ fixed-order chart's	≤ 0.64×	T18	Symptom-reading micro-F1 on a sealed bank (API key)	≥ 0.90 (Claude); ≥ 0.80 (rules)
T10	Confirmed-answer effort ÷ random probing's	≤ 0.59×	T19	Exact reading of ≥ 200 meter photos (API key)	≥ 95% (Claude); ≥ 90% (offline)
T11	Confirmed-answer effort, complaint read ÷ ignored	≤ 0.98×	T20	95th-percentile time per recommendation	≤ 0.5 s
T22	Out-of-model units flagged; separation (AUROC)	≥ 51% flagged; AUROC ≥ 0.77	T21	Mean API cost per diagnosis at list prices (API key)	≤ \$0.10

last changed; T18 uses a sealed bank), in three voices, never naming a part, each symptom dropped with probability 20% and others added with 2%. Stress sets: 500 units outside the catalog (double faults, wrong-value parts, solder bridges) and tolerances 1.5-3 times wider (a curve). Baselines: a fixed-order service chart (supplies, output, signal, DC, hum, response), half-split tracing (supplies, then halving the signal path with a tone, then DC at the first bad stage; the strongest script) and random probing (never unsolders) share the engine's inference and stopping rule, and the scripts unsolder their leading suspect at 50% belief; a language model, alone and with our simulator as a tool, picks from the same list on 300 stratified units and stops when it names an answer. For T7 a teammate who does not record inserts 60 faults in sealed random order into three 24 V line-driver boards. Study S1 will time at least six technicians with and without the tool, without a bar: six people cannot show a benefit reliably.

How the bars were set; no test unit has been run. A pilot simulated for this alone (200 channel-strip and 60 block units each, as-new and aged; 250 out-of-catalog), weighted like the test set, gave: right group first on 95.6% of as-new and 93.2% of aged units, a 2.4-point cost of aging (5.2% of aged units flagged “no single fault fits”, 1.6% misnamed). To a confirmed answer, the engine alone spent 16.0 units (6.1 probing, 9.9 unsoldering) and half-split tracing 24.6 (13.5 and 11.1). Reading the complaint moved accuracy from 94.3% to 95.6% and effort by -3.5%, so the reader must still prove itself (T11, T18). Each bar is the pilot estimate moved 1.5 standard errors against the system and rounded, never past the estimate; counting the pilot's own error, each primary then has a 59-69% chance of passing. T5 has no pilot without an API key: +10 points is the smallest margin we judged would justify a model per circuit over a general one. T7's 70% follows from what a pass must show: 49 of 60 is the smallest count whose exact lower bound clears it. Each primary is judged alone; if independent, all six pilot-estimable primaries would pass together with a chance near 6%, and about two are expected to miss; every miss will be reported.

Scope. In: single static faults (opens, shorts, drift, wear) and the hum and distortion they cause, in supplies, preamplifier and line-level stages. Out: power amplifiers, intermittent and mechanical faults (static simulation cannot produce them), double faults (flagged only), mains-side work, units without a schematic, unattended probing.

DESIGN DECISION · Score by ambiguity group and effort to a confirmed answer

Options: per-part accuracy and a count of measurements, or group accuracy and weighted effort to a confirmed answer.

Choice: the latter: per-part scoring rewards lucky guesses between faults with identical readings, unsoldering is not like touching a probe, and an answer is worth acting on only once confirmed. **Consequence:** group sizes (Table 2) and raw effort are reported too.

Part 4 — Preliminary AI approach, risks and ethics

Section lead: MEMBER_4

Why a hybrid. Language models guess when uncertain, partly because training and evaluation reward guessing [48]; the 97.59% system of Part 1 needed a simulation check. Simulation supplies the physics; Bayesian inference keeps a probability for every suspect group, updated after each reading through likelihoods learned from simulated units: one to three full-covariance Gaussians per fault (chosen by BIC) over the circuit's readings, giving a likelihood for any subset of readings in closed form. “No single fault fits” is a hypothesis with a broad flat density, calibrated on out-of-catalog units to a 5% false-alarm budget. Boosted trees [49] are a comparison engine (pilot top-1 91.8% against 94.3%). The next measurement maximizes expected information gain per unit of effort, the one-step entropy rule of model-based diagnosis [50] with a cost term (an optimal sequence needs a search [51]); a diagnosis stops at 90% belief in one group, at the budget, or when

Table 5. Initial risk register. Before/After: likelihood/impact, L, M, H or F (fatal: death or serious injury), reviewed each milestone; owners are the team’s leads. No risk with a fatal impact is accepted on procedural controls alone: it needs a lock in code or stays out of scope. The register uses the NIST AI RMF’s functions (govern, map, measure, manage) as a checklist, without claiming conformance [64].

Risk	Before	Control	Check	Owner	After
Shock; unqualified user	M/F	Hazard map; bring-up and discharge locks; license	T15, T16	Safety	L/F
Unit differs from its model	M/F	Checked against the chart, else every point hands-off	T13, T15	Simulation	L/F
Owner’s unit damaged	M/M	Owner approves removals; parts-removed log	T13, S1	Evaluation	L/M
Invented reading; injection; model change	M/H	Grounding check; confirmed photos; calibration; text read as data; model pinned	T12, T16, T17	Evaluation	L/M
Simulation-to-reality gap	H/H	Meter load, mains, aging; fault audit; boards	T6, T7	Simulation	M/H
Over-trust; deskilling; blame	M/H	Reasons, runners-up shown; trainee mode; sign-off limited	S1	Ethics	M/M
Work intensified; tickets used on staff	M/M	Roles, not names, on tickets; no staff rating	Interviews	Ethics	L/M
Privacy of complaints, photos	M/M	Offline default; metadata stripped; session-only data	T15-T19	Safety	L/M
Copied manuals, schematics	M/M	Shops model manuals they hold; models stay with them	License	Ethics	L/M

nothing informative remains. The language model (Anthropic’s Claude, version pinned) reads complaints (offline, rules do), proposes meter-photo readings for confirmation and explains; every number it writes must trace to a reading (T17), and it stays only if T5, T18 and T19 show it helps.

Set aside: consistency-based candidates [50] (tolerances blur conflicts); retrieval over manuals (they describe healthy units); hand-built troubleshooting networks (one per product, no ground truth); deep networks, which can condition on any subset of readings [52, 53] and choose measurements by information gain [54], but need more data and are harder to audit than one density per fault.

Autonomy. By the functions automated [55], we define five levels, named by analogy to driving automation: L0, the technician does everything; L1, the tool ranks suspects; L2, it also recommends each measurement; L3, it takes readings through instruments, a person approving each; L4, an unattended fixture. Differential sits at L2: a person chooses, takes and confirms every measurement, since a wrong step near high voltage can injure.

Safety. A fault can put high voltage where there is normally none: channel-strip point TP10, normally about 0 V, reaches 126 V under “C203 short circuit”. Steps at points that can exceed 50 V carry hands-off and discharge instructions; in a high-voltage chassis every powered step says “high voltage inside” and names a meter rated CAT II 600 V or better (IEC 61010), and powered steps wait until the technician records a bring-up through a variac or series lamp. Before a part is removed, the owner’s approval is recorded and the technician records under 2 V (our margin) at every point that can hold high voltage, which catches a capacitor left charged behind a failed-open resistor; the check lapses after 5 minutes without a bleeder, as a discharged capacitor recovers charge [30], and on any power-up. These are checks, not guarantees: readings are the technician’s attestation, and the hazard map covers only the modeled circuit; T15 checks every possible step.

People. Runner-up suspects and reasons are shown, yet AI can mislead: consultants given GPT-4 were 19 percentage points less likely to solve a task outside its competence [56], and those trusting AI more report less critical thinking [57]. Trainees commit their own next step before seeing the tool’s and need a named supervisor near high voltage; a tutor giving hints instead of answers largely mitigated the learning loss an unrestricted chat model caused [58]. S1 plants one wrong recommendation on a non-hazardous step of a 24 V board, after an ethics-board determination; participants are told beforehand that some advice may be wrong on purpose, are debriefed, and are recruited outside their employers. Tickets record roles, not names, and the license forbids rating staff; whether owners turn faster diagnosis into a faster pace is a question for the interviews. As fairness audits do for groups of people [59], errors will be broken down by complaint voice, fault type and circuit; a gap above 5 points on a primary blocks release until explained.

Accountability. The team answers for the hazard map, stated performance and corrections; a shop, for its circuit models and staff; the technician, for measurements and precautions. Whether advice software is a product (strict liability) or a service (negligence) is unsettled in the U.S., where bills would make AI systems products [60, 61] or shield developers when licensed professionals use them [62]; the revised EU directive treats software placed on the market after 9 December 2026 as a product [63].

DESIGN DECISION · Safety and security rules live in code, not only in the prompt

Options: instruct the model to be safe, or enforce safety in code. **Choice:** code: prompt injection ranks first in the OWASP Top 10 for LLM applications [65] and injected text can steer a model [66], so notes, complaints and photo text are data, the model’s tools act only on the session, and attestations come only from the technician’s own words. **Consequence:** warnings, refusals and locks behave the same in every mode; T15 checks them at every step.

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